

Splatt3R-SLAM: Feed-Forward Gaussian Splatting for Real-Time Dense SLAM, and an Honest Account of What It Costs

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Figure 1. **Novel-view rendering on Replica room0**, all three panels scored under one protocol (Sec. 5): a held-out frame that is a keyframe of neither system, rendered from the ground-truth pose Sim(3)-aligned into each map’s own frame, through one renderer and one metric implementation. The frame shown is the *median* of the per-frame Δ PSNR distribution, not a hand-picked one — an arbitrarily chosen frame on this scene can reverse the ordering (Sec. 6.4). Our map reaches this quality with 2.0M Gaussians against Photo-SLAM’s 89K; Sec. 7.1 shows that the size is load-bearing rather than redundant.

Abstract

Feed-forward two-view Gaussian predictors remove the per-scene optimisation that dense Gaussian-splatting SLAM normally requires, but a predictor trained on curated image pairs is not automatically a good map builder: a SLAM system fuses hundreds of keyframes over the same surfaces, a regime the pairwise training objective never observes. We integrate Splatt3R into MAST3R-SLAM and study what actually has to change for the fused map, rather than the individual prediction, to improve.

We report three results. First, adaptation must be confined to the Gaussian head: fine-tuning the frozen encoder with LoRA collapses reconstruction by 49%, while head-only training of the same budget improves deployed map quality on five dataset families. Second, a training-free lever — fading Gaussian opacity at map-assembly time in proportion to pointmap confidence — improves the online map in 10 of 12 cells with zero harmful cells, and we show why a training-time penalty cannot substitute for it. Third, we identify what opacity does in this architecture: it is a brightness-trim dial against the black-background ras-

teriser, not a confidence gate. A pre-registered opposite-sign prediction (opacity versus input luminance, negative for heads trained under synthetic darkening and positive for real-sensor heads) holds on 5/5 families, and unifies the first two results.

We then build all four comparison systems from source and evaluate every system through one protocol. Two findings follow that we believe matter beyond this system. (i) Published GS-SLAM rendering numbers and ours differ by roughly one protocol, not by quality: running Photo-SLAM ourselves yields 22.2 dB on Replica office0 where the literature reports ~ 30.9 dB for the same system, an offset independently reproduced as an 8.8 dB gap between rendering a map from its own estimated poses versus ground-truth poses. (ii) Our own headline advantage shrinks under scrutiny: on all eight Replica scenes we lead Photo-SLAM by +3.44 dB PSNR but win LPIPS on only 5/8, and a three-scene subset would have overstated the margin by 1.85 dB. Most importantly, our maps carry 25–100 $\times$ more primitives and 8–16 $\times$ more GPU memory, and pruning them to a baseline’s budget costs 8.7–10.3 dB — the map size is load-bearing, not padding. We report the compactness deficit as

a measured architectural limitation rather than a footnote.

1. Introduction

Dense visual SLAM has converged on 3D Gaussian splatting [7] as its map representation, because Gaussians render photorealistically in real time where neural fields [12] do not. The dominant designs — MonoGS [11], PhotoSLAM [5], SplaTAM [6] — obtain their Gaussians by *per-scene optimisation*: primitives are initialised, then densified and refined by gradient descent against the incoming frames.

A different route has become available. Feed-forward reconstruction networks predict geometry directly from images without optimisation: DUST3R [22] regresses pointmaps from an image pair, MAST3R [9] adds matching, and Splatt3R [16] extends MAST3R with a head that emits a full Gaussian per pixel — position, covariance, colour and opacity — in a single forward pass. MAST3R-SLAM [13] shows that the MAST3R prior supports real-time dense tracking. The combination is therefore attractive: a SLAM system whose map appears at the speed of a forward pass.

The difficulty is a mismatch of regimes. Splatt3R is trained on *curated image pairs*, and its loss sees exactly two views at a time. A SLAM system does something categorically different: it fuses hundreds of keyframes onto the same physical surfaces, so every surface is covered many times over, by predictions made from different viewpoints, at different exposures, with different confidences. Nothing in pairwise training observes this crowding. Our starting observation is that a checkpoint which is excellent per pair can nonetheless produce a fused map that is hazy and over-solid — and that fixing this is not a matter of training longer.

This paper asks what actually has to change, and reports the answer as three results.

Where adaptation is admissible. The obvious move — adapt the backbone to the target domain — fails decisively. Encoder LoRA [4] degrades reconstruction by 49% with a $1821\times$ scale explosion, and decoder-only LoRA improves for one epoch then collapses. The reason is structural: Splatt3R’s Gaussian head was trained against *frozen* encoder features, and perturbing those features feeds the head inputs it has never seen. Training only the Gaussian head, with the encoder bit-frozen, improves deployed map quality across five dataset families. Because the encoder is untouched, tracking is unchanged by construction — a property we verify externally rather than assume.

A training-free lever, and why training cannot replace it. Fading each Gaussian’s opacity at the moment it is injected into the map, in proportion to how little the pointmap vouches for it, improves the online map in 10 of 12 cells across four families, six heads and eight scenes, with two ties and *no* harmful cell. This is not a heuristic that a better loss would subsume: crowding is a map-level property that only exists after many keyframes accumulate, and a pair-training objective never sees a map. We support this with a dose–response experiment at two thinning depths.

What opacity is. Both preceding results turn out to be the same result. We show that opacity in this architecture is a *brightness-trim dial* against the black-background rasteriser, not the confidence or hedging channel it is usually assumed to be: because the colour term is anchored to the (possibly degraded) input pixel and is sigmoid-bounded, opacity is the only channel with headroom to brighten. The decisive evidence is a pre-registered opposite-sign prediction — heads trained under synthetic darkening should correlate opacity *negatively* with input luminance, real-sensor heads *positively* — which holds on all five families. Under this account the base checkpoint’s collapsed opacity means it *cannot* trim; injection-time thinning supplies the trim externally; and heads that deploy well are the ones that learned it.

An honest comparison, and what it cost us. We built MAST3R-SLAM, MonoGS, Photo-SLAM and VGGT-SLAM [10] from source and evaluated every system — ours included — through a single protocol, on held-out views that are keyframes of no system. Two findings emerged that we did not set out to make.

First, *published GS-SLAM rendering numbers are not comparable to ours*, and the gap is quantifiable. Running Photo-SLAM ourselves on Replica office0 gives 22.2 dB where third-party tables report ~ 30.9 dB for the same system. We independently reproduce an 8.8 dB gap on MonoGS between rendering its own map from its own estimated poses versus ground-truth poses. Two systems, two methods, the same ~ 8.7 dB: the literature and this paper differ by one protocol’s worth of evaluation choices.

Second, several of our own claims weakened when pushed. A three-scene Replica subset suggested a +5.29 dB advantage; completing all eight scenes reduced it to +3.44 dB and revealed that we win LPIPS on only 5/8. On TUM at matched compute we *tie* MonoGS on PSNR and *lose* on LPIPS. And our maps carry $25\text{--}100\times$ more Gaussians at $8\text{--}16\times$ the peak memory; pruned to a baseline’s budget our quality falls 8.7–10.3 dB *below* it, so the size is constitutive rather than redundant. We report these as measured results, with the curve, rather than as caveats.

Our contributions are: (i) a head-only adaptation recipe for feed-forward Gaussian SLAM, with the negative encoder-adaptation result that motivates it; (ii) an injection-time thinning lever with a dose–response argument for why training cannot substitute for it; (iii) a falsifiable mechanism for opacity, confirmed by a pre-registered opposite-sign prediction on 5/5 families; (iv) a single-protocol comparison against four locally-built systems, with a quantified protocol offset that we believe affects how this literature should be read; and (v) a measured account of the efficiency deficit our representation carries.

2. Related Work

Feed-forward 3D reconstruction. DUST3R [22] recasts two-view stereo as direct regression of aligned pointmaps, removing explicit correspondence search; MAST3R [9] adds a matching head and metric scale, and VGGT [21] generalises the formulation to many views. Splatt3R [16] builds on MAST3R by freezing the encoder and decoder and training only a Gaussian head that emits, per pixel, a covariance, spherical-harmonic colour and opacity (Eq. (1)–Eq. (4)). That freezing decision is not incidental for us: it is the reason the head is brittle to any change in its input features, which we quantify in Sec. 4.2. A second consequence matters equally: because Eq. (4) is masked, purely photometric and strictly pairwise, the training objective is blind to every property that only emerges when hundreds of keyframes are fused — which is precisely the regime a SLAM map occupies.

Gaussian-splatting SLAM. MonoGS [11] optimises Gaussians and camera poses jointly by gradient descent from monocular video. Photo-SLAM [5] couples ORB-SLAM3 [2] tracking with a Gaussian mapping thread. SplaTAM [6] and Gaussian-SLAM [23] take related routes with RGB-D input. All of these obtain map primitives by per-scene optimisation, and their map quality is therefore a function of the optimisation budget they are given — a dependency we make explicit when comparing (Sec. 6). Our system differs in that primitives arrive *predicted*: the optimiser refines an initialisation that is already approximately correct, rather than creating structure.

Feed-forward priors inside SLAM. DROID-SLAM [19] established learned dense optimisation for tracking. MAST3R-SLAM [13] uses the MAST3R prior for real-time dense tracking and pointmap fusion, and is the system we build on; we inherit its tracking front-end unmodified, which is why our trajectory accuracy matches it and why we are explicit that trajectory accuracy is *not* a contribution of this work. VGGT-SLAM [10] aligns VGGT submaps on the $SL(4)$ manifold; it produces poses and a

point cloud but no renderable map, so in our comparison it serves as a tracking control rather than a rendering baseline.

Parameter-efficient adaptation. LoRA [4] is the default tool for adapting a large frozen backbone. Our experience is a cautionary counterexample: for a pipeline whose downstream head was *trained against frozen features*, perturbing the backbone is not a mild intervention, and we measure a 49% collapse. The useful adaptation surface in such architectures is the head itself.

Evaluation practice. Rendering quality in this literature is reported as PSNR/LPIPS [24] and trajectory accuracy as ATE after Sim(3) alignment [20], typically via evo [3]. The details underneath those names vary substantially between papers — whether scoring views are training or held-out, whether rendering uses estimated or ground-truth poses, resolution, undistortion, and post-sequence optimisation budget. Section 5 shows that these choices are worth roughly 8.7 dB on unchanged systems, which is larger than most reported inter-method differences.

3. Preliminaries

3.1. Feed-forward pointmaps and Gaussians

DUST3R [22] maps an uncalibrated image pair $I^i, I^j \in \mathbb{R}^{H \times W \times 3}$ to pointmaps $X_i^i, X_i^j \in \mathbb{R}^{H \times W \times 3}$ with confidences C_i^i, C_i^j , where X_j^i denotes the pointmap of image i expressed in the frame of camera j . MAST3R [9] adds a matching head producing features $D_i^i, D_i^j \in \mathbb{R}^{H \times W \times d}$. We write $F_M(I^i, I^j)$ for one forward pass.

Splatt3R [16] adds a third head, in parallel to the pointmap and matching heads and using the same DPT decoder, which emits for every pixel a rotation quaternion $q \in \mathbb{R}^4$, a scale $s \in \mathbb{R}^3$, spherical-harmonic coefficients $S \in \mathbb{R}^{3 \times d}$, an opacity $\alpha \in \mathbb{R}$ and a mean offset $\Delta \in \mathbb{R}^3$, so that the Gaussian centre is

$$\mu = x + \Delta, \quad x \in X_i^i. \quad (1)$$

Quaternions are normalised, scales and offsets pass through an exponential and opacities through a sigmoid. Critically for this paper, *only the Gaussian head is trained*: the encoder and decoder remain those of the pre-trained MAST3R model, and all Gaussian parameters are predicted in the first image’s camera frame, avoiding any ground-truth transformation at inference.

The covariance follows the standard 3DGS [7] parameterisation

$$\Sigma = R(q) \text{diag}(s)^2 R(q)^\top, \quad (2)$$

and a view with world-to-camera W and projective Jacobian J renders with the projected covariance $\Sigma' =$

$JW\Sigma W^\top J^\top$, compositing front-to-back:

$$c = \sum_k c_k \alpha_k \prod_{l < k} (1 - \alpha_l). \quad (3)$$

Equation (3) is the reason opacity behaves as it does in our analysis (Sec. 4.5): with a black background, the residual transmittance $\prod_l (1 - \alpha_l)$ contributes zero radiance, so lowering α darkens a region and raising it brightens it.

Splatt3R is supervised only by novel-view renderings, under a mask M of pixels visible in at least one context view:

$$\mathcal{L} = \lambda_{\text{MSE}} \mathcal{L}_{\text{MSE}}(M \odot \hat{I}, M \odot I) + \lambda_{\text{LP}} \mathcal{L}_{\text{LPIPS}}(M \odot \hat{I}, M \odot I), \quad (4)$$

with $\lambda_{\text{LP}}=0.25$ in the released configuration. Two properties of Eq. (4) drive our method. It is *purely photometric* — depth enters only through the mask M and through pair selection, never through a gradient on geometry — and it is *pairwise*: the loss never observes more than two context views, hence never observes a fused map.

3.2. SLAM back-end

We inherit MAST3R-SLAM’s [13] front- and back-end unchanged. Poses are Sim(3) to absorb the scale inconsistency of network predictions,

$$T = \begin{bmatrix} sR & t \\ 0 & 1 \end{bmatrix}, \quad T \leftarrow \tau \oplus T \triangleq \text{Exp}(\tau) \circ T, \quad (5)$$

with $\tau \in \mathfrak{sim}(3)$. Under a generic central-camera assumption a pointmap induces its own camera model through the ray map $\psi(X_k^i)$ (unit-norm normalisation), and tracking minimises a ray error rather than a 3D point error, because $\|\psi_1 - \psi_2\|^2 = 2(1 - \cos \theta)$ is bounded and therefore robust to the depth errors that pointmap prediction frequently exhibits. The canonical keyframe pointmap is maintained by a confidence-weighted running average,

$$\tilde{X}_k^k \leftarrow \frac{\tilde{C}_k^k \tilde{X}_k^k + C_k^f T_{kf} X_k^f}{\tilde{C}_k^k + C_k^f}, \quad \tilde{C}_k^k \leftarrow \tilde{C}_k^k + C_k^f. \quad (6)$$

Loop closures enter a Sim(3) pose graph solved by Gauss–Newton. We modify none of this; the consequence is that our tracking is identical to MAST3R-SLAM by construction, which Sec. 6.6 verifies externally.

4. Method

4.1. System

We replace the pointmap prediction with Splatt3R, so each keyframe pair yields a per-pixel Gaussian in one forward pass. Keyframe Gaussians are stored in their anchor keyframe’s local frame and composed into the world through that keyframe’s *current* pose,

$$\mu^W = A_k \mu^k + b_k, \quad \Sigma^W = A_k \Sigma^k A_k^\top, \quad (7)$$

where $[A_k | b_k]$ is the Sim(3) matrix of keyframe k . Because the composition reads the pose at render time, a loop closure that updates T_{WC_k} moves the map with it at no cost and without re-baking — the property our refinement design depends on (Sec. 4.4).

4.2. Where adaptation is admissible

Splatt3R trains its head against *frozen* encoder features, making the head a function of one specific feature distribution. We measured what happens when that distribution moves:

- **Encoder LoRA** [4] ($r=8$, $\alpha=16$ on qkv/proj/fc1/fc2, 46.6M trainable): -49% reconstruction quality against the released checkpoint, with predicted scales exploding by $1821\times$. The failure is immediate, not a late-training instability.
- **Decoder-only LoRA**: $+0.37$ dB at epoch 1, then collapse to -1.61 dB by epoch 5 with a $12\times$ climb in the 99th-percentile scale.
- **Head-only** (encoder and decoder bit-frozen, only gaussian_dpt trainable): improves, and is what we keep.

The interpretation is structural rather than a matter of hyper-parameters. The scale head is exponentially parameterised, $s = \exp(\hat{s})$, so it is unconstrained above; feeding the head out-of-distribution features perturbs $\hat{s}$ in a regime where the photometric loss of Eq. (4) is nearly flat — a large, nearly transparent Gaussian and a small opaque one render almost identically at the supervision views. Geometry is therefore free to diverge while the loss barely moves, which is exactly the observed signature (quality collapse with scale explosion). Head-only training keeps 46.6M parameters frozen and trains the 158M-parameter head with the upstream objective and Adam [8], per dataset family.

4.3. Injection-time thinning

A SLAM map covers each surface with predictions from many keyframes. Under Eq. (3) with a black background, a surface covered by many over-solid Gaussians composites to haze rather than to a sharper surface. We therefore fade opacity at the moment a Gaussian enters the map, in proportion to how little the pointmap confidence vouches for it.

Confidences are not calibrated across keyframes, so we rank-normalise *within* each keyframe. For a keyframe with N Gaussians and confidence values $\{c_n\}$, let r_n be the rank of c_n and $\hat{c}_n = r_n/(N - 1) \in [0, 1]$. The injected opacity is

$$\alpha'_n = \alpha_n \cdot \text{clip}\left(1 - 2\lambda(1 - \hat{c}_n), 0.1, 1\right), \quad (8)$$

with dose $\lambda=0.45$ shipped. The lower clip at 0.1 is deliberate: it keeps Eq. (8) a *thinning* prior rather than a deletion

prior, so that even the least-confident Gaussian retains gradient and the refiner can recover it if the photometric evidence supports it.

Why a training-time penalty cannot substitute. An opacity penalty added to Eq. (4) would be a different intervention, because crowding *does not exist* at training time: it is a property of the assembled map, which forms only after SLAM accumulates keyframes over a surface, and Eq. (4) never sees more than two context views. Consistent with this, we measured that the refiner un-saturates opacity on its own even with an untrained head (median $1.0 \rightarrow 0.26$), identifying thinning as an *acceleration along a path the optimiser already walks* — which also explains its bounded downside, since a prior the optimiser would have reached anyway cannot do much harm. A dose-response check at two thinning depths (opacity knees 0.9 and 0.6) gave -21.0% and -15.8% LPIPS; neither collapsed toward zero, as a training-substitutable effect would require.

4.4. Online map refinement

Prediction gives a good initialisation, not a converged map. We add a third process that photometrically refines the assembled map *while SLAM runs*.

Parameterisation. Gaussians are optimised in their anchor keyframe’s local frame in pre-activation form — $\log s$, unnormalised q , $\logit \alpha$, and the SH DC term — so that scales stay positive and opacity stays in $(0, 1)$ without constraints, and composed to the world by Eq. (7) at every step. Per-group Adam rates follow the 3DGS reference, with the positional rate scaled by scene extent: $\eta_\mu = 1.6 \times 10^{-4}$, extent, $\eta_{f_{dc}} = 2.5 \times 10^{-3}$, $\eta_\alpha = 5 \times 10^{-2}$, $\eta_s = 5 \times 10^{-3}$, $\eta_q = 1 \times 10^{-3}$.

Objective. Supervision views are tracked frames, so the refiner optimises

$$\mathcal{L}_{\text{ref}} = (1 - \lambda_D) \|\hat{I} - I\|_1 + \lambda_D (1 - \text{SSIM}(\hat{I}, I)), \quad \lambda_D = 0.2, \quad (9)$$

i.e. the 3DGS objective rather than Eq. (4): LPIPS is a reporting metric here, not a training signal, so that our evaluation metric is not also our loss.

Anchor-carried supervision. Each supervision frame is stored as an image together with (anchor index, $T_{\text{anchor},f}$) rather than a world pose. Its world pose is recomposed through the anchor’s *current* pose at sample time, so supervision follows loop closures exactly as the map does under Eq. (7). Frames are held in CPU shared memory, which is what allows the refiner to run on a second GPU without pinning both processes to one card.

Sampling. Supervision is drawn from a reservoir of 200 frames spanning the whole sequence plus a ring of the 64 most recent, with a recent fraction of 0.3. We measured uniform-over-history to dominate a recency-weighted schedule (held-out early/late split: 14.40/14.09/14.72 uniform versus 14.06/13.79/14.35 recent-only), because causal arrival already biases the pool toward recent views.

Scheduling. Sharing one GPU, the refiner holds a duty cycle d by sleeping $\text{EMA}[\Delta t](1 - d)/d$ after each step, so optimisation steps are dropped but never frames. Refiner load on the *same* card doubles tracker latency ($101 \rightarrow 206$ ms p50); on a second card it is noise (103 ms), which is the configuration we deploy. After the sequence ends an optional polish phase continues with the final poses.

Frozen centres. We hold μ fixed during refinement. The centres come from the network’s pointmap, and photometric gradients degrade them: because the loss of Eq. (9) is evaluated at the supervision views themselves, moving a centre can reduce training error while worsening the geometry seen from new viewpoints. This was measured optimal at both online (~ 300 -step) and offline (3000-step) budgets.

4.5. What opacity is

The two preceding levers are, we argue, one phenomenon. Consider what the architecture permits. The DC colour is anchored to the input pixel plus a small learned residual and is sigmoid-bounded, so it cannot brighten a region much past what its input pixel suggests. Under Eq. (3) with a black background, α can: lowering it darkens toward black, raising it brightens by admitting less background. Opacity is therefore the only channel with headroom to *lift* content — a brightness-trim dial, not the confidence or hedging gate it is usually taken to be.

This makes a falsifiable, sign-specific prediction. Let ℓ be input luminance and ρ the Spearman correlation between α and ℓ over predicted Gaussians. A head trained with independently darkened inputs (synthetic vignetting and white-balance jitter, where the target view’s darkening cannot be inferred from the input) must learn to lift what the input darkened, so $\rho < 0$. A head trained on real sensor data with no synthetic darkening should instead place solid Gaussians on bright, reliably reconstructable content, so $\rho > 0$. We registered both signs before measuring; Sec. 6.3 reports the result.

5. Evaluation Protocol

Every number in this paper was produced by us on one machine ($2 \times \text{RTX A6000}$). All four comparison systems were built from source locally; *no number is copied from a paper*. This section states the protocol before the results, because

we will show that the protocol is worth more than most of the differences it is used to measure.

5.1. One protocol for every system

Rendering quality is scored identically for all systems:

- **Scoring views:** 100 held-out frames per scene, selected to be keyframes of *no* participating system, so every scored view is genuinely unseen by all of them.
- **Rendering pose:** ground-truth pose, Sim(3)-aligned [20] into each map’s own (arbitrary, monocular) frame.
- **Renderer and metrics:** one code path, one PSNR and one LPIPS [24] implementation, one exposure-normalisation setting, for every system.
- **Sensor parity:** monocular against monocular.

Trajectory accuracy is ATE RMSE via `evo_ape tum -as` [3].

Datasets. Rendering is evaluated on all eight Replica [17] scenes and on TUM RGB-D [18] `freiburg1`; trajectory accuracy on all nine TUM `freiburg1` sequences. Head-only adaptation (Sec. 6.1) additionally uses EuRoC [1], ETH3D [14] and 7-Scenes [15], giving five dataset families in total.

Alignment. Each monocular map lives in its own arbitrary Sim(3) frame, so ground-truth poses must be transported into it. We fit (s, R, t) by Umeyama’s closed-form solution [20] on the estimated/ground-truth camera-centre pairs and invert it,

$$R_{\text{map}} = R^\top R_{\text{gt}}, \quad t_{\text{map}} = R^\top (t_{\text{gt}} - t)/s, \quad (10)$$

noting that the $1/s$ belongs to the translation only — folding it into the rotation block yields a non-orthonormal matrix that silently corrupts every rendered viewpoint. Systems that log both estimated and ground-truth keyframe poses (MonoGS) are aligned from those pairs directly; the rest from their trajectory files.

5.2. A mandatory self-consistency check

Before any cross-system number is reported we require that *a map must render well from the poses it was built at*. This is not a formality. Our first MonoGS adapter assumed its logged trajectory was world-to-camera when it is camera-to-world; the inverted convention scored MonoGS at 4.36 dB against 18.71 dB for the correct one, and would have produced a large, flattering, publishable-looking margin. We adopted the rule that a result which favours us receives one more verification step than one that does not; it changed our conclusion three separate times.

5.3. The protocol offset

Two independent measurements quantify how much the protocol is worth on *unchanged* systems.

(i) Same system, our protocol versus the literature. Photo-SLAM built from its own source scores 22.23 dB on Replica office0 under our protocol; third-party comparison tables report ~ 30.9 dB for Photo-SLAM on monocular Replica. Offset: 8.7 dB.

(ii) Same map, two pose sources. Taking MonoGS’s own output map and changing only the rendering pose: 18.71 dB from its own estimated poses, 9.87 dB from ground-truth poses. Offset: 8.8 dB.

Two different systems, two different methods, essentially the same figure. We conclude that GS-SLAM rendering numbers in the literature and ours differ by approximately one protocol’s worth of evaluation choices — scoring on training versus held-out views, estimated versus ground-truth poses, resolution, undistortion, and post-sequence optimisation budget — rather than by quality. Concretely, our Replica 23.0 dB mean should *not* be read as trailing a published 30.9 dB: measured identically, we are ahead of the system that published it.

What the offset is made of. We can attribute part of it directly. Scoring the *same* map from estimated versus ground-truth poses is worth 8.8 dB on MonoGS and 2.6 dB on Photo-SLAM, the difference between the two reflecting how much each system’s map has co-adapted to its own pose error. Beyond pose source, the protocols we found differ in: whether keyframes are excluded from scoring (MonoGS excludes its own; Photo-SLAM’s self-reported PSNR is computed *on* its keyframes), resolution (640×480 undistorted versus our 512×384), masking (PSNR over $I_{\text{gt}} > 0$ pixels only), and post-sequence optimisation budget (26,000 iterations for MonoGS versus our 300 s polish). We do not claim to have decomposed the offset exactly; we claim that it is large, reproducible, and larger than most reported inter-method differences.

Reporting rule. We therefore tabulate only numbers we produced under one protocol. Published numbers are cited in text where relevant but are never placed in a comparison table beside our own. We recommend this practice generally for this literature.

5.4. Systems compared

Table 1 lists what each system can contribute. Two of the four produce no renderable map and therefore appear only in the trajectory table; MonoGS ships RGB-D-only Replica configurations, so including it there against our monocular system would give it a depth-input advantage and we omit it rather than report an unfair win.

System	Type	Rend.	Role here
Ours	mono 3DGS	✓	—
Photo-SLAM	mono 3DGS	✓	rendering
MonoGS	mono 3DGS	✓	rendering (TUM)
MASt3R-SLAM	mono dense	—	tracking control
VGGT-SLAM	pose+cloud	—	tracking control

Table 1. Systems compared: Photo-SLAM [5], MonoGS [11], MASt3R-SLAM [13], VGGT-SLAM [10]. The latter two produce no renderable map — we verified by source inspection that the VGGT-SLAM repository contains no rasteriser or photometric metric — so they appear only in the trajectory table. MASt3R-SLAM is the system we inherit tracking from, making it a control rather than an opponent.

6. Experiments

6.1. Head-only adaptation

Head-only training was run per dataset family and evaluated on the deployed map, not on validation pairs. Deployed LPIPS improves on four of five families: ETH3D -29.1% , EuRoC -20.8% , TUM -7.9% to -25.9% over four sequences, Replica positive, and 7-Scenes the weak family ($+0.14\%$ to -4.9%). The result is seed-robust: two heads $\times$ two seeds, checked at both validation and deployment level, gave relative deltas under 5% with inconsistent signs — ordinary noise rather than fragility.

Four falsified attributions. We attempted to explain why some families gain far more than others, and failed. Table 2 lists the candidates: each was pre-registered with the cell that would refute it, and each was refuted by direct measurement rather than by argument. We report this as a negative result. The gains are real — they exceed a measured 0.17% LPIPS noise floor — but we decline to adopt the most plausible surviving story in the absence of an out-of-sample test.

6.2. Injection-time thinning

Across 12 online cells spanning four families, six heads and eight scenes: 10 clear benefits, 2 ties, 0 harmful cells, worst cell $+0.16\%$ LPIPS. The confidence-weighted allocation of Eq. (8) is shipped as the default, though we note honestly that confidence-weighted versus uniform allocation is statistically a coin flip at this sample size (median -0.06% , CI containing zero); we ship it because it is never much worse and it is the configuration all deployment comparisons were run with.

Dose-response. If the effect were something a training-time penalty could supply, thinning should lose its benefit once the head already produces well-separated opacities. We measured two thinning depths on heads whose opacity

Candidate	Why it fails
Scale-frame consistency	Mechanically inert: Eq. (4) is purely photometric, means are <code>detach()</code> ed, and the metric-scale flag’s only consumer is disabled. Depth touches training solely via the mask.
Supervision coverage	Measured per family: EuRoC has the <i>lowest</i> coverage (39.5%) and the second-largest gain; ETH3D matches TUM’s coverage (61.9 vs 60.0%) with $3\times$ the gain.
Opacity un-saturation	Explains four families in rank order but 7-Scenes measures 0.479 — comparable to EuRoC’s 0.38 — with a null gain. A measured counterexample, not a pending cell.
Base-checkpoint headroom	$\rho = 0.145$ over 11 sequences. ETH3D <code>sofa_1</code> has the lowest deficit of any sequence and the second-largest gain.

Table 2. Four pre-registered mechanisms for the cross-family variation, each killed by measurement. The variation is real and remains unattributed.

Head	Training data	$\rho(\alpha, \text{luminance})$
photospatial	Replica + synthetic darkening	-0.316
mixed	Replica + synthetic darkening	-0.279
TUM	real sensor	$+0.303$
ETH3D	real sensor	$+0.392$
EuRoC	real sensor	$+0.304$

Table 3. Pre-registered opposite-sign prediction, confirmed on 5/5 families. Both signs were registered before any number was computed.

distributions differ, obtaining -21.0% and -15.8% LPIPS: neither collapsed toward zero. Together with the observation that the refiner un-saturates opacity unaided (median $1.0 \rightarrow 0.26$), this supports the reading of Eq. (8) as an acceleration of a trajectory the optimiser already follows, rather than as information the network was missing.

6.3. The opacity mechanism

Table 3 reports the pre-registered test. Heads trained under synthetic darkening correlate opacity *negatively* with input luminance; heads trained on real sensor data correlate *positively*. All five families land on the predicted sign, with magnitudes clustering tightly.

6.4. Rendering: Replica, all eight scenes

Table 4 is the load-bearing rendering comparison: monocular against monocular, on the benchmark this literature actually uses for rendering.

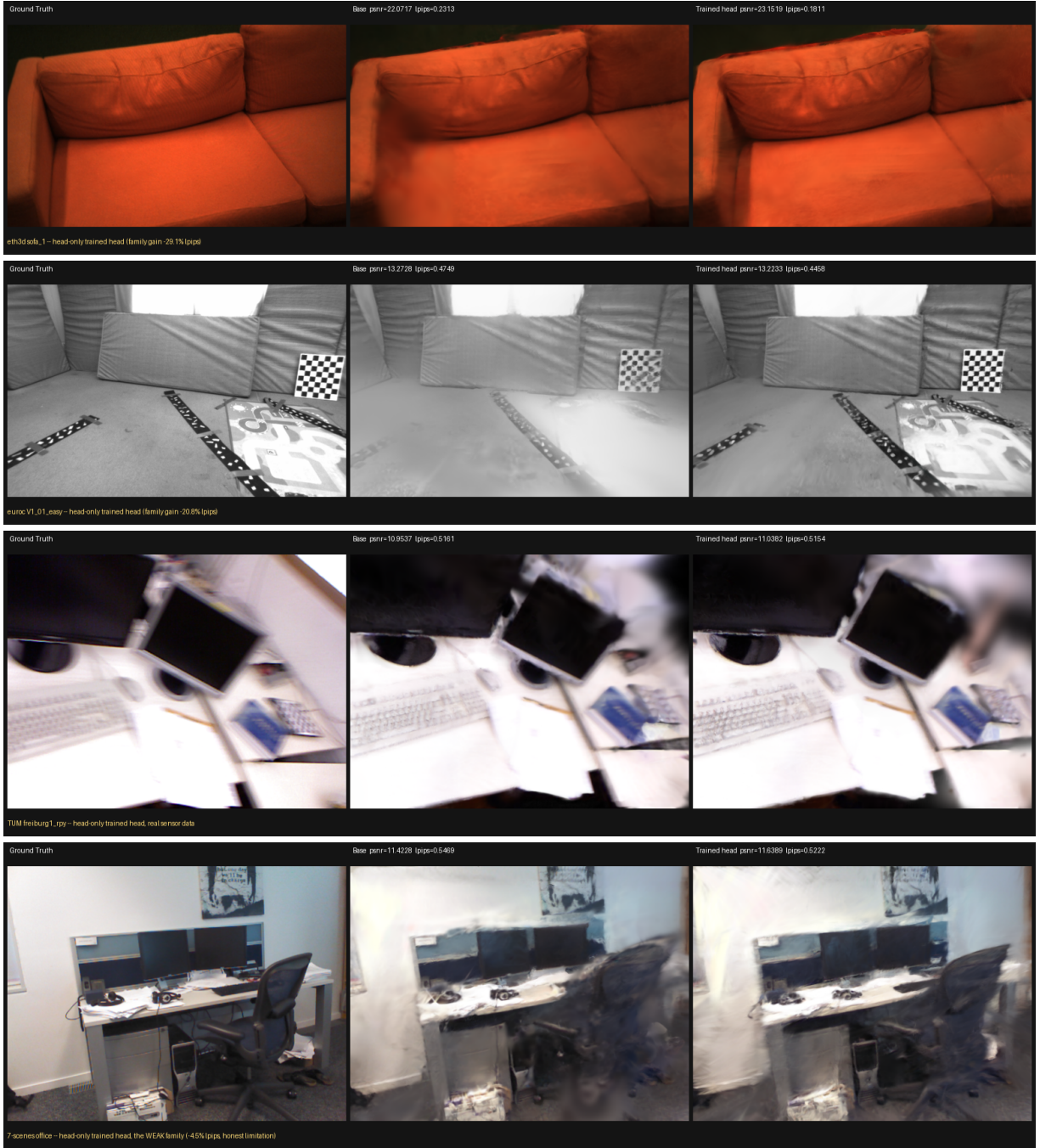


Figure 2. **Head-only adaptation, per family** (ground truth | released checkpoint | head-only fine-tuned), on held-out non-keyframe views. Rows: ETH3D sofa_1, EuRoC V1.01, TUM fr1_rpy, 7-Scenes office. The first three are families where the recipe clearly helps; the last is the weak family (+0.14% to -4.9% LPIPS), included so the failure mode is visible rather than only tabulated. Because the encoder is bit-frozen, the difference between the second and third panel of each row is attributable to the Gaussian head alone.

Means hide the distribution. Scoring 24 probe frames per scene individually, the per-frame Δ PSNR median is +8.01 on room0 and +3.62 on office0, but -1.84 on office2 — a scene whose *mean* says we lead by +0.73 dB. Our nominal win there comes from a minority of frames,

not from being typically better. Even on room0, PhotoSLAM wins some frames (min $\Delta = -4.77$). Qualitative figures in this paper are therefore selected at the median of the per-frame delta, not by eye.



Figure 3. **Replica office0, one protocol.** A second baseline comparison under the identical protocol used for Fig. 1, on a scene where our margin is closer to the Replica mean (+4.08 dB versus the +3.44 dB eight-scene mean) rather than at the top of the range.

Scene	Ours		Photo-SLAM	
	PSNR $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	LPIPS $\downarrow$
office0	26.30	0.104	22.23	0.209
office1	22.08	0.115	17.80	0.185
office2	20.97	0.151	20.24	0.129
office3	20.18	0.144	19.36	0.155
office4	23.65	0.156	16.46	0.125
room0	25.44	0.110	17.92	0.152
room1	21.91	0.140	21.88	0.175
room2	23.62	0.160	20.75	0.116
Mean	23.02	0.135	19.58	0.156

Table 4. Replica, monocular, one protocol. We lead PSNR on 8/8 but LPIPS on only 5/8. A three-scene subset (office0, office1, room0) would have reported +5.29 dB; all eight give +3.44 dB, and three scenes are effectively ties (+0.03, +0.73, +0.81).

System	PSNR $\uparrow$	LPIPS $\downarrow$	Gauss.	Budget
Ours (300 s)	13.95	0.411	2.40M	~ 1.4 k st.
MonoGS (339 s)	13.87	0.398	23K	26k it.
Photo-SLAM	10.73	0.558	36K	to shutdown
Ours (1200 s)	14.61	0.379	2.40M	~ 4.4 k st.

Table 5. TUM `fr1_desk`. This comparison cannot be extended to further TUM sequences: MonoGS’s monocular tracking diverges on them (Tab. 6), and scoring rendering against a diverged map would measure tracking failure rather than rendering quality.

6.5. Rendering: TUM, and the budget question

Table 5 reports TUM `fr1_desk`, the only TUM sequence where all three rendering systems track successfully. At *matched wall clock* — our 300 s polish against MonoGS’s 339 s refinement — PSNR ties and MonoGS wins LPIPS. Only with $3.5\times$ their budget do we lead both. The honest statement is that MonoGS has better perceptual quality at equal compute and we can buy past it with more, not that our deficit was a budget artefact.

Seq.	Ours	MASt3R	VGGT	Photo	MonoGS
360	0.042	0.048	0.050	0.035	0.177
desk	0.017	0.016	0.025	0.015	0.036
desk2	0.028	0.024	0.029	0.438	0.844
floor	0.027	0.025	0.099	0.013	0.539
plant	0.015	0.020	0.025	0.046	0.071
room	0.059	0.061	0.064	0.510	0.791
rpy	0.022	0.023	0.026	0.057	0.041
teddy	0.048	0.045	0.036	0.305	0.123
xyz	0.009	0.009	0.014	0.010	0.017
Mean	0.030	0.030	0.041	0.161	0.293

Table 6. ATE RMSE (m), TUM freiburg1, all nine sequences and all five systems. Photo-SLAM is non-deterministic (ORB-SLAM3 is multi-threaded): `desk` gave 0.0174 and 0.0149 on two runs, so its single-run figures carry run-to-run variance.

6.6. Trajectory accuracy

Table 6 completes the trajectory picture. The finding is *robustness*, not precision: Photo-SLAM and MonoGS each diverge on several sequences (0.30–0.84 m, i.e. failure rather than error), which is what dominates their means, while on sequences they do track Photo-SLAM is frequently the most accurate. We stress that our parity with MASt3R-SLAM (4–4–1, noise) is *expected and not a contribution*: the tracking front-end is inherited unchanged, which Tab. 6 verifies externally rather than by assertion.

6.7. Refinement ablations

Table 7 isolates the components of Sec. 4.4. The refiner’s own contribution, measured with the base checkpoint on both arms so that only the refiner toggles, is large and consistent across families: +8.69 dB on ETH3D, +4.62 dB on Replica, +3.52 dB on TUM and +1.00 dB on EuRoC. This is not a marginal add-on next to head-only training — on ETH3D it exceeds it — and the two levers stack rather than compete.

Ablation	Δ PSNR	Note
Refiner on, ETH3D	+8.69	300 s polish
Refiner on, Replica	+4.62	
Refiner on, TUM	+3.52	
Refiner on, EuRoC	+1.00	
Causal replay, 120 it.	+1.21	strict real time
Second GPU, unthrottled	+2.15	latency +1%
Uniform vs. recency	+0.34	held-out mean
Voxel dedup, 10 mm	-0.53	-27.6% map size

Table 7. Refinement ablations. ATE is unchanged in every row (0.016975 vs. a 0.0170 baseline in the deterministic configuration): the map is strictly downstream of tracking, and the pre-registered failure condition “ATE moves at all” never fired. Freezing the Gaussian centres was separately measured optimal at both the online (~ 300 -step) and offline (3000-step) budgets. The dedup row is included as a negative: it is a map-size control that costs quality when it fires without a recovery budget, not a quality feature.

6.8. System cost

On TUM `fr1_desk`, one GPU, one sampler for all systems: ours 70 s wall clock at 21,035 MiB peak; Photo-SLAM 28 s at 1,286 MiB; MonoGS 555 s at 2,389 MiB; VGGT-SLAM 24 s at 9,436 MiB. Our peak memory is $16\times$ Photo-SLAM’s and $8.8\times$ MonoGS’s, and Photo-SLAM is $2.5\times$ faster end-to-end.

7. Limitations

7.1. The map is not compact, and the size is load-bearing

Our maps carry $25\text{--}100\times$ more primitives than the GS-SLAM baselines at $8\text{--}16\times$ the peak GPU memory. The natural defence would be that this is redundancy — prune it and parity remains. We tested that defence and it is false.

Table 8 truncates our own refined maps to successively smaller budgets by keeping the highest-opacity Gaussians (the ones the optimiser is itself fading are the honest ones to drop first), re-scoring on the identical held-out frames with no re-optimisation afterwards.

We state the bound in both directions. This is naive post-hoc truncation without re-optimisation, so every row is a *lower* bound — a system actually trained or refined at that budget would do meaningfully better. But the caveat does not rescue the defence: the measurement shows our quality is *constituted* by having millions of primitives rather than padded by them. Training or refining directly at a small budget, or a densification-control policy, is future work we have not done.

Gaussians kept	office0	office1	room0	room2
full ($\sim 2\text{--}3\text{M}$)	26.29	22.08	25.44	23.62
1,000,000	24.68	21.71	25.01	22.87
300,000	20.23	18.96	16.26	17.87
83,372	13.54	13.97	8.68	10.43
23,019	12.04	12.08	5.16	7.74
<i>Photo-SLAM at $\sim 83K$</i>	<i>22.23</i>	<i>17.80</i>	<i>17.92</i>	<i>20.75</i>

Table 8. PSNR versus map budget, four Replica scenes. At Photo-SLAM’s own primitive budget we are 8.7/3.8/9.2/10.3 dB *worse* than it, having been 4.08/4.28/7.53/2.87 dB better at $25\text{--}36\times$ the count. The knee sits consistently near 1M Gaussians.

7.2. Scope of the comparison

Rendering on TUM is $n=1$ and cannot be extended, because MonoGS’s monocular tracking diverges on the remaining sequences. Optimisation budgets are matched by wall clock on TUM but not on Replica, where Photo-SLAM runs to its own shutdown iteration. Resolution and undistortion handling still differ per system and we have not quantified their residual effect. Photo-SLAM was built against OpenCV 4.10 rather than the 4.7/4.8 its authors tested, because our CUDA 12.4 requirement forces it; this touches its image front-end but not its Gaussian mapping.

7.3. Unresolved threads

Three questions we could not close, stated rather than omitted. The cross-family variation in head-only gains survived four falsified attributions and remains unexplained. Cross-frame colour consistency is diagnosed but not fixed — our photospatial augmentation happens to reduce hue disagreement by 12.6% on the sequence that originally exposed the bug, which is a side-effect rather than a solution, since the single-view objective contains nothing that penalises cross-view inconsistency. Refitting the loop-closure retrieval assets on Splatt3R features looked favourable offline (Recall@1 +20% relative) but regressed catastrophically on one of three sequences in real SLAM, and was rejected — one more instance of an offline metric win that did not survive the online check.

8. Conclusion

We integrated a feed-forward Gaussian predictor into a dense SLAM system and asked what has to change for the *fused map*, rather than the individual prediction, to improve. Adaptation must be confined to the Gaussian head: perturbing a backbone whose downstream head was trained against frozen features collapses reconstruction by 49%, whereas head-only training helps on five families. A training-free lever — fading opacity at injection in proportion to pointmap confidence — helps in 10 of 12 online cells

with no harmful cell, and cannot be replaced by a training-time penalty because crowding is a map-level property that pair training never observes. Both results are explained by one mechanism: opacity in this architecture is a brightness-trim dial against the black-background rasteriser, confirmed by a pre-registered opposite-sign prediction that held on 5/5 families.

Evaluated through a single protocol against four locally-built systems, our method leads Photo-SLAM on all eight Replica scenes in PSNR (+3.44 dB mean) while winning LPIPS on 5/8, ties MonoGS on TUM at matched compute while losing LPIPS there, and inherits MAST3R-SLAM-class tracking robustness that the two GS-SLAM baselines do not match. It does so with a map 25–100× larger, and we show that this size is constitutive rather than redundant.

Two methodological points seem to us more transferable than the system itself. First, GS-SLAM rendering numbers in the literature and ours differ by approximately 8.7 dB of *protocol* — measured twice, on two systems, by two independent methods — so cross-paper rendering comparisons in this area should be treated as unreliable unless the protocols are stated and matched. Second, in the course of this work three separate results that initially favoured us changed when checked one level deeper: an inverted pose convention, a misread self-consistency test, and a small-scene subset that overstated our margin by 1.85 dB. We adopted the rule that a result which flatters the authors earns one more verification step than one that does not, and we recommend it.

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